Off-Road Trafficability Assessment With Remote Sensing Imagery and Incomplete Auxiliary Data via a Cross-Modal Channel Feature Fusion Network

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***Abstract*—Off-road trafficability (ORT) in complex geological environments is crucial for special operations, emergency rescue, and natural resource development. ORT is influenced by a combination of geographical and geological environmental factors. Recent studies primarily employ rule methods for ORT assessment. These approaches are labor-intensive, lack timeliness and objectivity, and are constrained to limited data sources. To address these issues, this work proposes a new method for ORT assessment that combines remote sensing imagery with geographic and geological data (RSI-factors) through a cross-modal rectified fusion network (CRFNet). This network is designed with the feature rectification module (FRM) and feature fusion mixer module (FFMM) to integrate cross-modal features. To address potential missing data in complex geological environments, this work proposes a multitask and prompt learning strategy (MtPLS) to improve model robustness. Experiments conducted on our Asia Dataset and Africa Dataset yielded optimal evaluation results. On the complete Asia and Africa Dataset, the CRFNet model improved overall accuracy (OA) by more than 25% and the kappa coefficient by over 43% compared to rule methods. On the incomplete Asia and Africa Dataset, the CRFNet model improved OA by more than 5.5% and the kappa coefficient by over 8% compared to suboptimal deep learning (DL) models. To the best of our knowledge, this research work is the first in which DL features have been combined with multimodal RSI-factors data for ORT assessment, paving a new path for research in this field. The source codes of this work will be made publicly available at https://github.com/kangkanghe/CRFNet**

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***Index Terms*—Deep learning (DL), geographic and geological data, off-road trafficability (ORT) assessment, remote sensing.**

I. Introduction

OFF-ROAD trafficability (ORT) refers to special operations in which personnel and equipment can be safely and efficiently transported to a predetermined target location in complex geological environments lacking infrastructure such as roads and bridges [1], [2]. These operations are influenced by more than ten geological environmental factors such as elevation, slope, soil moisture, land cover (LC), and others [3]. These factors exert varying degrees of influence on ORT depending on spatiotemporal conditions, a phenomenon referred to as spatiotemporal heterogeneity. To address this variability and establish a standardized ORT evaluation framework, researchers commonly represent ORT assessments as classification maps that delineate different trafficability conditions, "good," "restricted," "engineering measures required," and "difficult" [4]. Accurately identifying and assessing ORT is a significant scientific issue in fields such as special operations, emergency rescue, and natural resource development [5], [6].

Currently, researchers primarily focused on use manual rules to assess ORT, using relevant remote sensing data and auxiliary data, such as LC, the soil remold cone index (RCI), and slope [4], [7], [8], [9], [10], [11]. This manual rule method, based on the experience of geological or military experts, is labor-intensive, lacks timeliness and objectivity, and is limited to specific data sources. Moreover, with the increase in the number of assessment factors, the complexity of manually formulating ORT rules escalates significantly. To address this challenge, researchers have proposed automated quantitative rule method (QRM) that consider complex geological environmental factors, such as geological hazards (GHs), slope, and LC, initially achieving automated ORT assessment [3]. However, expert experience is still required as an aid. Consequently, there is an urgent necessity to develop intelligent methods for the rapid assessment of ORT in complex geological environments.

Deep learning (DL) and other artificial intelligence technologies have become mainstream methods for image classification [12], [13], [14], [15], [16], [17], [18]. In ORT assessment, it is necessary to comprehensively consider remote

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sensing imagery with geographic and geological environmental factors (RSI-factors). There are significant challenges in developing a unified model to integrate RSI-factors data. Existing multimodal fusion methods are mainly divided into two categories: input-stage fusion, which directly combines different data sources [19], [20], [21] and deep feature fusion, which achieves integration through multilevel interactions [22], [23], [24]. Each fusion paradigm has its advantages, but the optimal fusion strategy usually depends on the specific dataset and classification task, and currently, no universally applicable fusion strategy exists [25]. In this work, we are dedicated to developing a unified RSI-factors image classification model to enable rapid, accurate, and efficient ORT assessment. However, The multimodal fusion mechanism heavily rely on the completeness of the data, leading to high application barriers for multimodal RSI-factors data. ORT assessment targets areas of interest to special operations, emergency rescue, and natural resource development in complex geological environments. These areas are typically denial zones (areas explicitly prohibited for entry by personnel or vehicles for security and strategic reasons) [26]. While geographic data on denial zones are widely available, geological data [such as rock and soil (RS) and joint structure (JS)] typically rely on "manual visual interpretation and field investigation," making data completeness difficult to ensure [27]. Ensuring model robustness when multimodal data are missing is a challenge [25].

To address the aforementioned challenges, this work proposes a novel cross-modal rectified fusion network (CRFNet) that leverages the unique characteristics of RSI-factors data and integrates the strengths of existing fusion paradigms. Drawing inspiration from the works of Pundir, Garg et al., and He et al., we structure ORT assessments into three categories: good going (I), restricted going (II), and difficult going (III), as illustrated in Fig. 1. This network employs a dual-stream architecture consisting of an RSI feature stream and a factors feature stream, and it incorporates the feature rectification module (FRM) and the feature fusion mixer module (FFMM). Factors data exhibit variations in temporal and spatial resolution, spectral characteristics, and spatiotemporal heterogeneity across different terrain conditions. The FRM module rectifies multimodal features using spatial, channel, and interaction correlations, reducing data uncertainty and noise. In addition, traditional ORT assessments primarily rely on geographic data and rarely exploit the high-level semantic features of RSI. The FFMM module integrates RSI image features with factors features, achieving cross-modal fusion of multimodal data. This optimizes traditional methods and effectively extracts critical information from RSI-factors imagery. Furthermore, this work proposes a multitask and prompt learning strategy (MtPLS) to improve the model's robustness in the face of missing data. At a glance, the following contributions are presented.

1) For the first time, DL features are combined with RSI-factors multimodal data for ORT assessment in complex geological environments, while the CRFNet model was developed to integrate RSI-factors data for efficient ORT assessment.

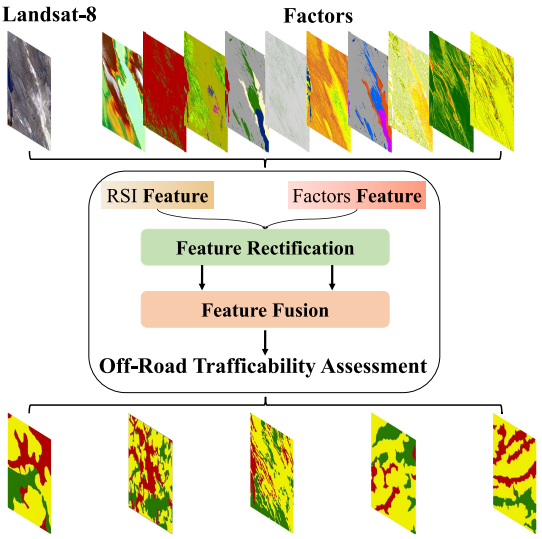


Fig. 1. ORT assessment process based on the CRFNet model assigns three categories: green, indicating "good going" (I); yellow, representing "restricted going" (II); and red, signifying "difficult going" (III).

2) The FRM module addresses differences in spatial and temporal resolution, spectral characteristics, and other aspects of multimodal data, thereby reducing the interference of data uncertainty and noise on ORT assessment.
3) We propose the MtPLS to improve the robustness of the CRFNet model when dealing with missing modal data due to privacy or security restrictions in complex geological environments.
4) We selected two regions with typical geological and topographical conditions in Africa and Asia to create the Asia Dataset and Africa Dataset. Extensive experiments have demonstrated the superiority of the CRFNet network.

II. Related Work

A. ORT Assessment

Initially, researchers primarily focused on the impact of single factors on ORT assessment, finding certain limitations in its application [28], [29]. Subsequently, Pundir and Garg et al. [8] conducted field experiments to investigate the potential of multiple factors, such as LC and soil moisture, in ORT assessment, finding that the combination of multiple factors significantly improved classification accuracy. Since then, researchers have widely used RSI and related auxiliary data for ORT assessment. Pundir and Garg et al. [4] proposed a rule-based method using data such as LC, soil, DEM, and moisture to assess ORT. In further research, Pundir and Garg et al. [30] developed an empirical relationship model for assessing soil spatial variability through terrain similarity analysis and geospatial data, with which they assessed the impediments of terrain, soil, and moisture on ORT. These studies effectively applied multiple factors in ORT assessment. In addition, other researchers have employed multicriteria decision analysis methods to assess the influence of various geographic and geological environmental factors on ORT in

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special operations [7]. These studies have primarily relied on expert experience to exhaustively enumerate all possible trafficability rules in a study area, then used manual rules to constrain soil types, moisture, slope, and other data for GIS raster overlay analysis to generate ORT assessment maps. Subsequently, He et al. [3] devised a QRM that comprehensively assesses the impact of both geographic and geological environmental factors on ORT, achieving automated assessment. However, this method still requires manual determination of factor weights and information on factor correlations in different study areas. In addition, RSI has been seldom used in previous research. In this work, RSI is integrated with supplementary information from more than 10 geographic and geological environmental factors to improve ORT assessment.

B. Multimodal Remote Sensing Image Classification

Although rule-based methods have achieved certain successes in ORT assessment, they rely on significant human labor and lack timeliness and objectivity. Therefore, there is an urgent need for more intelligent and efficient ORT assessment methods. In recent years, the development of DL image classification technologies has provided new solutions to this problem [31], [32]. Ewing et al. [33] utilized thermal imaging, multispectral data, and hyperspectral data, combined with machine learning and DL algorithms, to predict soil moisture and terrain strength, and they analyzed the maneuverability of the Polaris MRZR vehicle. Zhang et al. [34] classified terrain using DEM and backpropagation neural networks. These methods not only improved classification accuracy but also reduced human intervention, demonstrating the significant potential of DL in ORT assessment.

The challenge of using DL methods for ORT assessment lies in the feature fusion of more than ten types of multimodal RSI-factors data. Currently, research on multimodal RSI classification has been effectively applied in hyperspectral image classification [28], [35], [36], [37]. The mainstream paradigm is to develop fusion schemes to bridge multiple parallel modal streams [38], [39], [40], [41], [42]. Fan et al. [43] demonstrated the powerful potential of multimodal data fusion in urban village classification and geospatial analysis by using a multilayer feature fusion network to integrate remote imagery and street view images. Yao et al. [36] enhanced the vision transformer with novel features, such as separable convolution and cross-modality attention, for improved land use and LC classification, outperforming traditional models on multimodal datasets. Roy et al. [35] developed the MFT using attention across multiple patches and LiDAR to enhance HSI LC classification, achieving superior results. In this work, we advocate a similar paradigm that aims to propose a unified framework for addressing the RSI-factors ORT assessment problem in complex geological environments without the need for terrain-specific adjustments.

Although multimodal RSI-factors data have achieved optimal results in ORT assessment, their fusion mechanisms heavily rely on the completeness of the modalities. Therefore, another important direction in multimodal fusion is to develop models with high robustness to incomplete multimodal data [44]. Ma et al. [25] studied the robustness of multimodal Transformers to missing modalities and improved their fusion strategy through multitask optimization to handle different missing modality scenarios. Lee et al. [45] proposed a multimodal prompting technique that effectively performs visual recognition even with partially missing modalities. These works explored different strategies for robust adaptability to the issue of missing modalities, prompting the integration of various strategies into ORT assessment under the general scenario of missing multimodal RSI.

III. Methodology

ORT assessment is influenced by a combination of Landsat-8 imagery and various geographical and geological environmental factors, such as DEM, LC, RS, GHs, JS, terrain wetness index (TWI), RCI, stream power index (SPI), topographic position index (TPI), and slope. For this task, the present work proposes the CRFNet, as shown in Fig. 2. The main workflow of this network can be divided into four stages: RSI-factors feature extraction, FRM (see Section III-B), FFMM (see Section III-C), and MtPLS (see Section III-D).

A. Architecture Overview

The architecture of CRFNet is presented in Fig. 2(a). It utilizes two parallel branches: the RSI images branch and the factors images branch, to extract features from Landsat-8 data and multimodal factors data, respectively. By leveraging complementary information from these data sources, CRFNet enhances the accuracy of ORT assessment. However, due to the inherent heterogeneity of multimodal data across temporal, spatial, and semantic dimensions, direct fusion may result in semantic misalignment or amplified noise. To address this issue, CRFNet introduces an FRM, as shown in Fig. 2(b), which aligns the feature spaces of RSI and Factors branches prior to fusion. Unlike traditional attention mechanisms that focus solely on spatial or channel domains, FRM enforces intermodal spatial-semantic consistency before feature interaction, thereby enhancing the robustness of cross-modal representations. Subsequently, an FFMM is designed to exchange and integrate rectified features from both the RSI and factors branches, as depicted in Fig. 2(c). In contrast to conventional Mixer structures, FFMM explicitly preserves the semantic structure of each modality branch during fusion, improving the effectiveness and discriminability of cross-modal feature integration. Finally, a MtPLS is incorporated to generate modality-incomplete data during training. By combining input-level prompt injection with a multitask learning paradigm, MtPLS enhances the model's performance under missing modality conditions. Notably, this strategy requires minimal architectural modifications while significantly improving the robustness of CRFNet. CRFNet adopts ResNet50 as the image modality backbone due to its proven stability and strong feature extraction capability in remote sensing tasks. It provides a good tradeoff between accuracy and computational cost, which is critical for real-time ORT assessment. Compared to Transformer-based backbones, ResNet50 achieves comparable performance with lower inference overhead in multimodal fusion scenarios.

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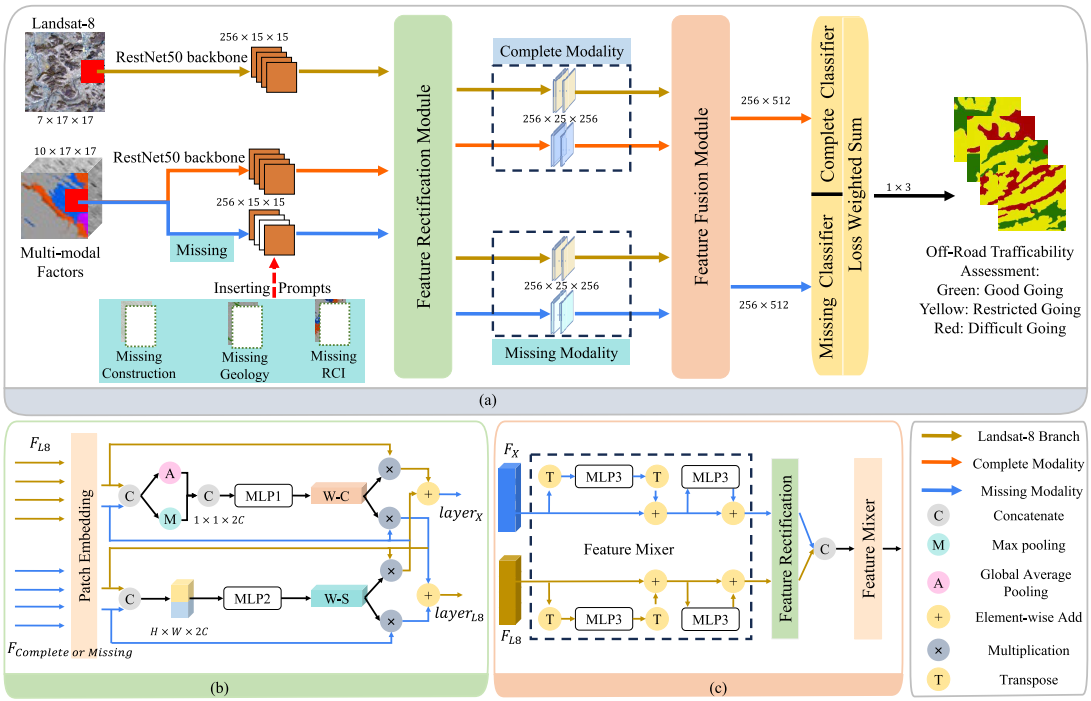


Fig. 2. (a) Architecture of the proposed CRFNet. (b) FRM is specifically designed for RSI-factors images originating from different sensing modalities. (c) FFMM is used for the deep fusion of RSI-factors features.

B. Feature Rectification Module

As analyzed above, RSI-factors images originating from different sensing modalities often offer complementary information. However, these images are also accompanied by uncertainties and noise. To mitigate the noise effects associated with these diverse modal features, we propose a novel FRM, as illustrated in Fig. 2(b). The FRM incorporates both channel-wise and spatial-wise rectification strategies, enabling more precise feature refinement and enhancing the reliability of ORT assessments.

The FRM, inspired by the convolutional block attention module [46], employs channel-wise and spatial-wise attention mechanisms to process heterogeneous modality features from Landsat-8 RSI branches and multimodal Factors image branches. Specifically, the input features from $\text{RSI}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$ and $\text{Factors}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$ are refined to enhance modality-specific representations. The FRM integrates two complementary rectification strategies: channel-wise feature rectification and spatial-wise feature rectification. In the channel-wise feature rectification process, the input features are concatenated along the channel dimension to combine complementary information from both modalities. Global max pooling and global average pooling are then applied to extract key statistical representations. The pooled features are concatenated into a vector $Y \in R^{2C}$, which is passed through a lightweight multilayer perceptron (MLP), denoted as F_{MLP1}. The MLP output is activated using the Sigmoid function, reshaped, and permuted to generate the channel-wise weights W^C_{RSI} and W^C_{Factors}, ensuring adaptive feature weighting for each modality. Finally, these weights are applied to rectify the input features RSI_{in} and $\text{Factors}_{\text{in}}$, producing the rectified features $\text{RSI}^C_{\text{rec}}$ and $\text{Factors}^C_{\text{rec}}$

$$W^C_{\text{RSI}}, W^C_{\text{Factors}} = F_{\text{reshape}}\left(F_{\text{MLP1}}\left(Y\right)\right) \tag{1}$$

$$\text{RSI}^C_{\text{rec}} = \lambda_c W^C_{\text{Factors}} * \text{Factors}_{\text{in}} \tag{2}$$

$$\text{Factors}^C_{\text{rec}} = \lambda_c W^C_{\text{RSI}} * \text{RSI}_{\text{in}} \tag{3}$$

where * denotes element-wise multiplication, and λ_c is a hyperparameter.

The input for spatial-wise feature rectification is also RSI_{in} and $\text{Factors}_{\text{in}}$, concatenated along the channel dimension. However, unlike the channel-wise approach, this step focuses on optimizing local features. The concatenated features are processed by F_{MLP2}, which consists of two 1×1 convolutional layers, with ReLU and Sigmoid activation functions interleaved. This design ensures efficient mapping from high-dimensional to low-dimensional space. After reshaping and permuting, the output generates spatial-wise weights W^S_{RSI} and W^S_{Factors}. These weights are used to rectify the input features, resulting in the rectified features $\text{RSI}^S_{\text{rec}}$ and $\text{Factors}^S_{\text{rec}}$

$$W^S_{\text{RSI}}, W^S_{\text{Factors}} = F_{\text{reshape}}\left(F_{\text{MLP2}}\left(\text{RSI}_{\text{in}} \| \text{Factors}_{\text{in}}\right)\right) \tag{4}$$

$$\text{RSI}^S_{\text{rec}} = \lambda_s W^S_{\text{Factors}} * \text{Factors}_{\text{in}} \tag{5}$$

$$\text{Factors}^S_{\text{rec}} = \lambda_s W^S_{\text{RSI}} * \text{RSI}_{\text{in}} \tag{6}$$

where λ_s is a hyperparameter.

Finally, the input features RSI_{in} and $\text{Factors}_{\text{in}}$ are sequentially processed through channel-wise feature rectification and spatial-wise feature rectification to generate the rectified features RSI_{rec} and $\text{Factors}_{\text{rec}}$. These refined features are subsequently fed into the next stage for feature fusion

$$\text{RSI}_{\text{rec}} = \text{RSI}_{\text{in}} + \text{RSI}^C_{\text{rec}} + \text{RSI}^S_{\text{rec}} \tag{7}$$

$$\text{Factors}_{\text{rec}} = \text{Factors}_{\text{in}} + \text{Factors}^C_{\text{rec}} + \text{Factors}^S_{\text{rec}}. \tag{8}$$

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C. Feature Fusion Mixer Module

The FFMM, inspired by the mixer architecture [47], is designed to process modality-specific features independently. It includes two parallel and independent Mixer branches to optimize the input features from $\mathrm{RSI}_{\mathrm{rec}} \in \mathbb{R}^{B\times P\times C}$ and $\mathrm{Factors}_{\mathrm{rec}} \in \mathbb{R}^{B\times P\times C}$, respectively. The feature shape for both inputs is $(256 \times H \times W)$. These inputs are transformed into a higher dimensional feature space of shape $(p \times h)$ using two patch embedding layers, where p denotes the number of image patches and h represents the embedding dimension. Each image patch is processed by a 2-D convolutional layer configured with an input channel of 256, an output channel of h, and both the kernel size and stride set to $64/\sqrt{p}$. With p and h set to 16 and 256, respectively, this process converts each patch into an h-dimensional vector, producing patch-embedded features for the RSI and Factors inputs. The FFMM module then utilizes mixer layers and skip connections to effectively fuse the RSI and factors feature maps. Two sets of feature mixers are applied to enhance cross-modal feature interaction, ensuring the generation of robust fused features.

Each mixer layer is composed of two components: a token-mixing MLP and a channel-mixing MLP. The MLP, denoted as F_{MLP3}, consists of two fully connected layers interleaved with a GELU activation layer. The token-mixing MLP treats the feature columns of each channel as independent inputs and achieves feature fusion by modeling relationships between different spatial locations (tokens). This process begins with applying layer normalization to the input features, followed by transposing the features. The transposed features are then processed using F_{MLP3}, and subsequently transposed back, producing high-level features with enhanced discriminative power. In contrast, the channel-mixing MLP treats the feature rows of each token as independent inputs, focusing on capturing interdependencies between channels to enable interactive feature fusion. This module is constructed using layer normalization and F_{MLP3}, effectively modeling dependencies between channels. Taking $\mathrm{RSI}_{\mathrm{rec}}$ as an example, the computation process of the feature mixer is outlined as follows:

$$\mathrm{RSI}_{\mathrm{mix}} = \mathrm{RSI}_{\mathrm{rec}} + F_{\mathrm{MPL3}}\left(\mathrm{GLUE}\left(\mathrm{Norm}\left(\mathrm{RSI}_{\mathrm{rec}}\right)\right)\right). \quad (9)$$

In the feature mixer, the computation processes for $\mathrm{RSI}_{\mathrm{rec}}$ and $\mathrm{Factors}_{\mathrm{rec}}$ are identical. After passing through the feature mixer, the resulting features are denoted as $\mathrm{RSI}_{\mathrm{mix}}$ and $\mathrm{Factors}_{\mathrm{mix}}$. These features are concatenated along the channel dimension and further refined through a second feature mixer to enhance feature interaction. The concatenated feature block undergoes layer normalization and is processed using global average pooling, which improves information interaction and extracts high-level semantic features. In the MLP-mixer, the patch size is set to 3, while the fully connected layers in the mixer layers are configured with 256 neurons. The token-mixing MLP contains 128 neurons, whereas the channel-mixing MLP is configured with 1024 neurons to capture dependencies more effectively. Finally, the fused features are fed into the classifier. After being processed through the FFMM module, two types of fused features are generated: F_{complete} and F_{missing}. These features are subsequently input into the multitask learning stage for joint optimization, enabling the model to effectively handle both complete and missing modality scenarios.

In ORT assessment scenarios, training data is typically modality complete, while testing data often suffers from modality incomplete issues. To address this challenge, we actively simulate modality incomplete data during the training phase to enhance the model's robustness. During training, the FFMM processes input features in two modes: a combination of $\mathrm{RSI}_{\mathrm{mix}}$ and $\mathrm{Factors}_{\mathrm{mix}}$, or a combination of $\mathrm{RSI}_{\mathrm{mix}}$ and $\mathrm{Missing}_{\mathrm{mix}}$. The computation process and dimensionality of $\mathrm{Missing}_{\mathrm{mix}}$ are designed to be identical to those of $\mathrm{Factors}_{\mathrm{mix}}$, ensuring compatibility and consistent feature representation across modalities.

D. Multitask and Prompt Learning Strategy

In this work, we address modality missing scenarios commonly encountered in practical applications. While most data within the multimodal factors is sourced from global public datasets, geological data (e.g., RS, JS, and RCI) often depends on manual interpretation, making it challenging to ensure completeness. To mitigate this issue, we introduce the MtPLS. As illustrated in Fig. 2(a), MtPLS simulates real-world scenarios during the training phase by generating modality incomplete data inputs. Specific positional prompt vectors are initialized using a normal distribution and combined with complete data to produce two types of losses: one corresponding to complete modalities and the other to missing modalities. These losses are weighted and summed to optimize the model. In the testing phase, MtPLS generates only the positional prompt vectors for the missing modalities, omitting the complete modality loss to align with real-world conditions. The total loss function is defined as follows:

$$L = \lambda_{\mathrm{com}} L_{\mathrm{com}} + \lambda_{\mathrm{mis}} L_{\mathrm{mis}} \quad (10)$$

where L_{com} represents the loss of the complete dataset; L_{mis} represents the loss of the incomplete dataset; and λ_{com} and λ_{mis} are hyperparameters that balance each loss, with $\lambda_{\mathrm{com}} + \lambda_{\mathrm{mis}} = 1$.

Prompt learning is an effective method for transfer learning in natural language processing. Based on the insertion point, it is divided into input-level prompts, attention-level prompts, and output-level prompts, with input-level prompts being the simplest and most efficient method [33]. The input features of our "generated" incomplete modality data are denoted as $h^i \in \mathbb{R}^{L\times d}$, where $i = 1, 2, \ldots, N$ represents the channels of the input feature vectors, and L and d represent the shapes of the input feature vectors. The prompt features designed for the missing RS, JS, and RCI data are denoted as $P^m \in \mathbb{R}^{L_p\times d}$, where L_p and d represent the shapes of the prompt feature vectors, and $m \in \{\mathrm{RS}, \mathrm{JS}, \mathrm{RCI}\}$ indicates the different missing modality factors. The final prompt function can be written as

$$F_{\mathrm{pro}}\left(P^m, h^i\right) = \left[P^m; h^i\right] \quad (11)$$

where $[\cdots;\cdots]$ represents the concatenation operation.

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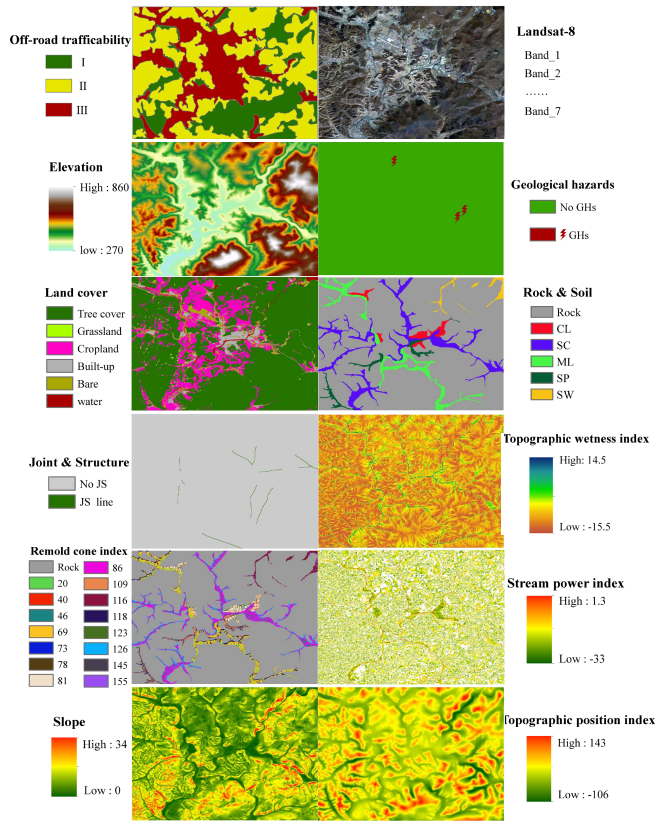


Fig. 3. Asia Dataset includes these labels and RSI-factors: Landsat-8, elevation, GHs, LC, RC, JS, TWI, RCI, SPI, slope, and TPI.

IV. Experimental Results and Analysis

A. Datasets

In this work, we constructed two datasets: the Asia Dataset and the Africa Dataset, both of which we used to test and validate the performance of the CRFNet. These datasets underwent projection transformation and resampling to standardize into the (GCS_WGS_1984) coordinate system with a spatial resolution of (9.55 m). Each dataset contains various types of maps: Landsat-8 imagery, DEM maps, GHs susceptibility distribution maps or GHs maps, LC maps, RS distribution maps, JS maps, TWI maps, RCI maps, SPI maps, slope maps, and TPI maps, as shown in Figs. 3 and 4.

This work employed Google Earth Engine to acquire Landsat-8 and DEM imagery. From the DEM data, slope, TPI, SPI, and TWI were derived. In addition, GHs maps were created through manual field investigations, identifying three critical hazard points. LC imagery was obtained from the ESA WorldCover dataset. RS and JS data were collected using a combination of "manual visual interpretation and field investigation." The RCI is influenced by multiple parameters, including soil type, thickness, compaction, moisture content, liquid limit, plastic limit, and compressive modulus, making large-scale quantitative evaluation challenging. To address this complexity, this work referred to the field manual for terrain Analysis (FM5-33) of the U.S. Army, summarized in Table I, to simplify RCI estimation. RCI values were computed using RS data and TWI, and further refined with field investigation data to achieve higher accuracy. The vehicle cone index (VCI),

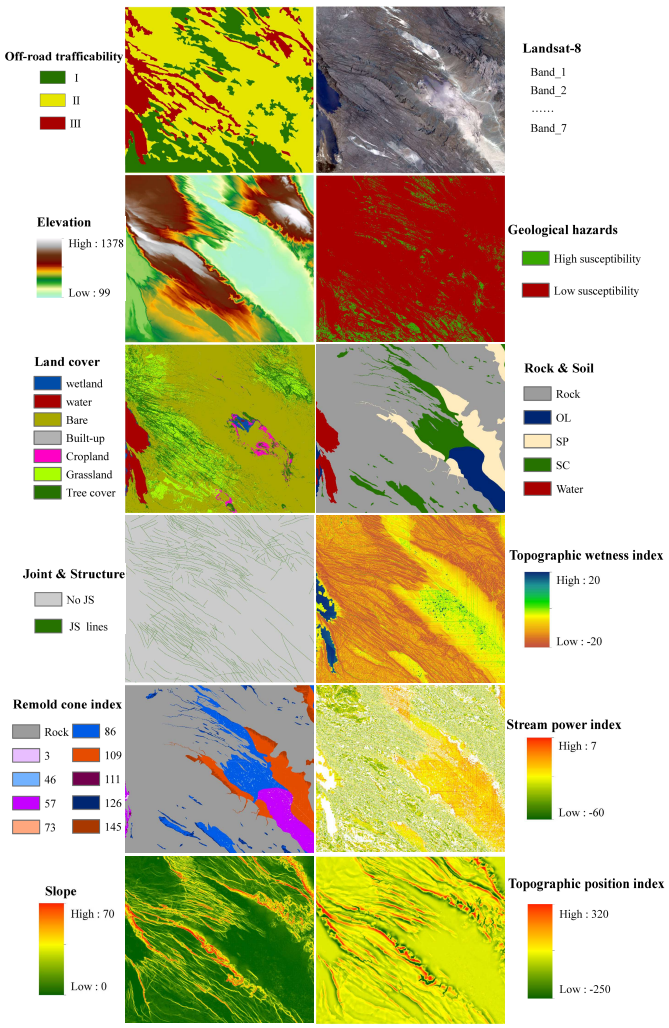


Fig. 4. Africa Dataset includes these labels and RSI-factors: Landsat-8, elevation, GHs, LC, RC, JS, TWI, RCI, SPI, slope, and TPI.

also derived from FM5-33, represents the cone index specific to vehicles. For instance, for the M1A1 Abrams main battle tank, VCI1 is 25 and VCI50 is 58. When RCI > VCI, the ground conditions are sufficient to support vehicle traffic. Typically, ORT assessments involve multiple categories (>2), such as "I," "II," and "III." For tracked vehicles, the ORT of the Asia dataset was categorized into three classes (I, II, and III). This dataset comprises 728 × 483 pixels. Landsat-8 imagery includes 7 bands, while Factors imagery incorporates 10 bands. Table II provides a detailed summary of the label categories and sample sizes.

1) Asia Dataset: The Asia Dataset covers a geologically complex mountainous and hilly area, primarily composed of metamorphic and sedimentary rock formations, with quaternary sediments present in some regions. The diverse terrain and various geological characteristics significantly impact ORT, making it ideal for studying ORT assessments. Due to the geological and topographical complexity of the area, manually creating an accurate ORT label dataset based on over ten RSI-factors is nearly impossible. Therefore, we utilized the QRM proposed to generate preliminary ORT assessment results for the region [3]. Due to significant noise in the

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TABLE I
RCI VALUES FOR DIFFERENT SOIL TYPES AND MOISTURE CONDITIONS, SOURCED FROM THE U.S. ARMY FIELD MANUAL FOR TERRAIN ANALYSIS (FM5-33)

Soil category	Description	RCI Values		
		Dry	Moist	Wet
GW	Gravel or sandy gravel, well graded	163	123	83
GP	Gravel or sandy gravel, poorly graded	160	120	81
GM	Gravel, silty	120	76	32
GC	Gravel or sand gravel, clayed	130	91	52
SW	Sand, well graded	155	116	78
SP	Sand, poorly graded	145	109	73
SM	Sand, silty	119	72	25
SC	Sand, clayey	126	86	46
ML	Silts	118	69	20
CL	Clays	123	81	40
OL	Organic silts	111	57	3
MH	Inorganic elastic silts	114	61	8
CH	Fat clays	136	99	62
OH	Fat organic clays	107	54	1
PT	High organic soils or peat	106	52	0
R	Rock outcrops	165	165	165
W	Open water	0	0	0

TABLE II
DESCRIPTION OF THE ASIA DATASET AND AFRICA DATASET, INCLUDING ORT TYPES AND THE NUMBER OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Asia Dataset		Africa Dataset	
		Training	Testing	Training	Testing
1	Good	891	88,298	4,396	435,245
2	Restricted	1,620	160,393	10,634	1,052,856
3	Difficult	803	79,565	2,631	260,488
Total		3,314	328,256	17,661	1,748,589

assessment results, we conducted noise filtering, boundary smoothing, and manual calibration, resulting in the ORT label data shown in Fig. 3. The Asia dataset was validated using a combination of visual remote sensing interpretation and field investigation to establish the ORT label dataset. This region encompasses four survey routes, with more than 200 validation points collected to ensure data accuracy and reliability.

2) Africa Dataset: The Africa Dataset covers an area of the Republic of Djibouti, where the surface is predominantly covered by cenozoic volcanic rocks and quaternary sediments. The terrain is mainly characterized by plateaus and plains, and seismic activity is frequent. He et al. [3] applied QRM to assess the ORT of plains, plateaus, and mountainous terrains in this region. Based on the ORT assessment results obtained through QRM, we performed noise filtering, boundary smoothing, and manual calibration to generate the ORT label data shown in Fig. 4. The Africa dataset was also validated using a combination of visual remote sensing interpretation and field investigation to establish the ORT label dataset. This region covers two survey routes, with more than 100 validation points collected to ensure data accuracy and reliability. The Africa Dataset was also validated using a combination of visual remote sensing interpretation and field investigation to establish the ORT label dataset. This region covers two survey routes, with more than 100 validation points collected to ensure data accuracy and reliability. The Landsat-8, DEM, LC, RS, JS, TWI, RCI, SPI, slope, and TPI image-acquisition methods are the same as those used for Fig. 3. The susceptibility of the study area to debris flows and landslides was assessed using CNN methods, generating GHs images that categorized the area into two classes: potential hazard areas and safe areas. The Asia Dataset contains images of 1429×1266 pixels; detailed statistical data are shown in Table II.

B. Experimental Setup

To validate the effectiveness of the CRFNet model, an extensive series of experiments were conducted to compare the model against traditional methods and state-of-the-art DL network models. The comparison methods included the following: 1) traditional manual rule [4]and QRM [3]; 2) the classic CNN model, specifically the two-branch CNN [48]; 3) several advanced feature fusion models, such as an extended vision transformer (ExViT) [36], a multilevel spatial-channel feature fusion (FusionMixer) [43], and a multimodal fusion transformer (MFT) [35]; and 4) advanced feature fusion networks for incomplete data, such as missing aware prompts (MAPs) [45]. Specifically, the manual rule method combined three types of multimodal data: RC, slope, and TWI [4]. The QRM utilized ten types of multimodal data: DEM, GHs, RS, LC, JS, SPI, slope, TWI, TPI, and RCI [3]. The two-branch CNN method conducted experiments using three different data settings: RSI, factors, and RSI-factors [48]. The feature fusion models FusionMixer, MFT, and ExViT were all used to conduct experiments using RSI-factors.

The training of the CRFNet utilized the Adam optimizer with a learning rate of 0.001, a batch size of 256, and spanned 100 epochs. All code implementations of the model were based on Pytorch (version 1.11.0), Torchvision (version 0.12.0), and Python (version 3.9.18). All experiments were conducted on a desktop equipped with 64 GB of memory and an NVIDIA RTX 3090 GPU. To accurately assess ORT, two widely recognized quantitative metrics were used to measure the model's performance and accuracy in different scenarios: overall accuracy (OA) and the kappa coefficient.

C. Performance of ORT Assessment

The experiments were divided into two major categories: complete modal datasets and incomplete modal datasets. Each experiment was conducted three times to confirm result reliability, with mean values and standard deviations reported. In practical applications, the Asia Dataset used 17×17 image patches, while the Africa Dataset used 19×19 image patches.

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TABLE III
OA (%) AND KAPPA (%) COEFFICIENTS RESULTS FOR THE COMPARISON METHODS ON THE COMPLETE ASIA DATASET, WITH THE OPTIMAL RESULTS HIGHLIGHTED IN BOLD

No.	Manual rule	Quantitative rule	Two-branch CNN			FusionMixer	MFT	ExViT	CRFNet
	Factors		RSI	Factors	RSI factors	RSI factors			
1	44.17	43.08	75.63 ± 1.32	79.67 ± 3.90	84.38 ± 1.76	84.95 ± 4.70	86.24 ± 2.98	85.72 ± 1.54	**91.61 ± 0.62**
2	56.71	68.77	66.76 ± 1.11	76.11 ± 3.69	79.39 ± 0.20	84.93 ± 1.40	82.86 ± 1.58	83.20 ± 2.11	**89.78 ± 0.57**
3	5.58	69.67	50.22 ± 0.97	66.36 ± 3.14	61.62 ± 3.01	74.64 ± 8.21	63.32 ± 1.72	71.95 ± 2.96	**84.94 ± 0.39**
OA	42.85	62.25	65.13 ± 0.12	72.83 ± 0.29	76.42 ± 0.82	82.44 ± 0.54	79.04 ± 0.40	81.15 ± 0.04	**89.10 ± 0.05**
kappa	2.81	38.96	44.34 ± 0.03	56.42 ± 0.57	62.20 ± 1.45	71.95 ± 1.03	66.28 ± 0.52	69.93 ± 0.30	**82.69 ± 0.08**

The hyperparameters λ_c and λ_s, used to balance channel-wise and spatial-wise effects, respectively, were both set to the default value of 0.5. In addition, the hyperparameters L_{com} and L_{mis}, used to balance the loss between complete and incomplete datasets, were both set to the default value of 0.5. A detailed discussion on the parameters and computational complexity of using the CRFNet for the datasets is presented in Section IV-C3.

1) Results for Asia Dataset: Table III presents a quantitative comparison of the proposed CRFNet model against other methods in terms of OA and the Kappa coefficient, with the optimal results highlighted in bold. The results demonstrate that quantitative rule-based methods outperform traditional manual rule-based methods, with OA and Kappa coefficients improving by 19.4% and 36.15%, respectively. This improvement arises from the ability of quantitative rule-based methods to consider over ten geographic and geological environmental factors, accurately capturing terrain complexity and significantly enhancing the stability and consistency of evaluations. In contrast, traditional manual rule-based methods rely heavily on expert experience and incorporate only a limited number of geographic factors, leading to insufficient generalization in complex and diverse scenarios.

When using the same factors data, CNN models achieve considerably higher accuracy in ORT assessment compared to quantitative rule-based methods, with OA and Kappa improving by 10.58% and 17.46%, respectively. This improvement highlights the limitations of quantitative rule-based methods, which rely on pixel-wise overlay analysis and fail to effectively capture local dependencies and global contextual relationships in spatial data, resulting in inferior performance compared to DL models. The two-branch CNN model, which integrates RSI-factors data, achieves an additional improvement in OA and Kappa by 3.59% and 5.78%, respectively. This enhancement is attributed to the incorporation of RSI data, which provides rich complementary information for ORT assessment. Moreover, the model's independent feature extraction branches for multimodal data effectively capture modality-specific features. As the depth of feature fusion increases, the model further improves accuracy, with OA and Kappa increasing by over 1%. The FusionMixer model achieves hierarchical feature interaction through MixerBlocks, enabling token and channel mixing to fully fuse multimodal features and effectively handle partially missing modalities. The MFT model incorporates cross-attention modules to enhance intermodal interaction, significantly improving the representation of both local and global features. The ExViT model combines deep hierarchical convolution with Transformer-based structures, achieving a balance between local texture and global relationship modeling.

Among all the compared methods, the CRFNet model achieves the optimal performance, with average OA and Kappa values of 89.1% and 82.69%, and standard deviations of 0.05% and 0.08%, respectively. Compared to the second-best FusionMixer model, CRFNet improves OA and Kappa by 6.66% and 10.74%, respectively. These superior results can be attributed to CRFNet's ability to dynamically rectify intermodal features using channel weights and spatial weights, effectively mitigating noise caused by differences in spatial and temporal resolution, spectral characteristics, and spatiotemporal heterogeneity in multimodal data. In addition, the token-mixing MLP and channel-mixing MLP enable deep fusion of RSI-Factors features, significantly enhancing the model's adaptability to complex multimodal scenarios.

We conducted a further analysis on the performance of the aforementioned DL methods on the incomplete Asia Dataset. All experimental data consisted of RSI-factors, with the test data masked for RS, GHs, and RCI. All models were retrained, and the results are presented in Table IV, with the optimal results highlighted in bold. The OA and kappa coefficients for all methods significantly decreased when multimodal data were missing. The CNN model shows the lowest accuracy, as the two-branch CNN performs feature fusion through simple concatenation in the final stage, failing to adequately capture intermodal interactions. This limitation leads to a decline in performance when handling complex multimodal data with missing features. The ExViT model exhibits the most severe decline, with OA and Kappa coefficients decreasing by 14.75% and 26.61%, respectively, followed by the MFT model. This is because MFT and ExViT rely on multilayer attention mechanisms and complex feature extraction strategies when processing data. However, they lack specific compensation modules for missing modalities, resulting in substantial performance degradation when intermodal interaction information is lost. Among all compared models, MAPs achieved the highest performance, with OA and Kappa coefficients of 81.36% and 70.81%, respectively. This improvement can be attributed to MAPs' ability to dynamically adjust feature representations in response to missing modalities, enabling the model to make effective inferences even with incomplete data and thereby mitigating the impact of modality loss. In contrast, the CRFNet model demonstrated optimal performance in scenarios with

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TABLE IV

OA (%) AND KAPPA (%) COEFFICIENTS RESULTS OF THE COMPARISON METHODS ON THE INCOMPLETE ASIA DATASET, WITH THE OPTIMAL RESULTS HIGHLIGHTED IN BOLD

No.	FusionMixer	MFT	ExViT	MAPs	CRFNet	CRFNet (single modality missing)		
						RS	RCI	GHs
1	36.42 ± 3.96	30.58 ± 6.54	31.37 ± 3.52	73.59 ± 3.52	**81.94 ± 1.08**	**92.03**	90.42	89.43
2	**92.71 ± 0.53**	81.13 ± 3.43	83.30 ± 2.86	91.69 ± 3.52	90.34 ± 0.40	88.89	**90.98**	89.46
3	81.29 ± 1.60	67.90 ± 1.00	71.23 ± 2.25	64.51 ± 3.52	**85.65 ± 0.70**	83.84	80.52	**86.73**
OA	74.80 ± 0.19	64.33 ± 0.35	66.40 ± 0.08	81.36 ± 0.24	**87.16 ± 0.15**	88.52	88.36	**88.79**
Kappa	57.34 ± 0.32	39.77 ± 1.06	43.32 ± 0.75	70.81 ± 0.52	**79.45 ± 0.27**	81.79	81.28	**82.22**

TABLE V

OA (%) AND KAPPA (%) ACCURACY RESULTS OF THE COMPARISON METHODS ON THE COMPLETE AFRICA DATASET, WITH THE OPTIMAL RESULTS HIGHLIGHTED IN BOLD

No.	Manual rule	Quantitative rule	Two-branch CNN			FusionMixer	MFT	ExViT	CRFNet
	Factors		RSI	Factors	RSI factors	RSI factors			
1	70.18	34.20	37.51 ± 2.17	90.16 ± 1.46	89.22 ± 0.69	92.99 ± 0.88	89.54 ± 0.94	91.37 ± 0.98	**93.06 ± 0.39**
2	56.70	77.52	86.21 ± 1.05	91.28 ± 1.44	92.75 ± 0.34	92.68 ± 0.93	93.07 ± 0.24	93.31 ± 0.13	**94.51 ± 0.46**
3	2.17	79.57	50.04 ± 1.01	77.35 ± 5.37	76.39 ± 0.75	83.19 ± 3.34	78.77 ± 0.21	78.28 ± 0.52	**83.71 ± 1.05**
OA	57.84	67.01	68.72 ± 0.06	88.93 ± 0.02	89.43 ± 0.13	91.34 ± 0.12	90.07 ± 0.21	91.05 ± 0.27	**92.54 ± 0.09**
Kappa	13.48	40.91	37.79 ± 0.81	79.94 ± 0.28	80.68 ± 0.21	84.40 ± 0.33	81.87 ± 0.44	81.77 ± 0.53	**86.46 ± 0.15**

TABLE VI

OA (%) AND KAPPA (%) ACCURACY RESULTS OF THE COMPARISON METHODS ON THE INCOMPLETE AFRICA DATASET, WITH THE OPTIMAL RESULTS HIGHLIGHTED IN BOLD

No.	FusionMixer	MFT	ExViT	MAPs	CRFNet	CRFNet (single modality missing)		
						RS	RCI	GHs
1	80.06 ± 3.36	79.41 ± 2.60	83.77 ± 1.66	84.12 ± 2.23	**91.26 ± 1.04**	90.01	92.08	**92.53**
2	91.47 ± 3.75	91.12 ± 1.51	87.27 ± 5.40	92.84 ± 3.31	**95.31 ± 0.14**	**95.70**	93.43	95.06
3	67.82 ± 2.27	62.75 ± 5.18	62.95 ± 14.24	64.90 ± 2.47	**77.48 ± 1.78**	80.70	**87.01**	81.20
OA	85.11 ± 1.20	83.99 ± 0.51	82.79 ± 1.82	86.23 ± 1.23	**91.95 ± 0.10**	92.06	92.15	**92.38**
Kappa	72.49 ± 1.48	70.05 ± 1.45	68.67 ± 2.76	74.83 ± 2.14	**85.34 ± 0.21**	85.37	85.87	**86.06**

missing multimodal data, with average OA and kappa coefficients of 87.16% and 79.45%, respectively, representing decreases of 1.94% and 3.24%. Among all the compared methods, the accuracy of Label I drops significantly, indicating that RS, GHs, and RCI play a critical role in determining this category in the Asia Dataset. To further investigate the impact of missing modalities, we conducted experiments analyzing the effect of individual modality removal on model performance. The absence of the GHs modality has the least impact on overall performance, primarily due to its high redundancy, as illustrated in Fig. 3. In contrast, the absence of either the RS or RCI modality results in a more noticeable decline in performance, suggesting that these two modalities are more essential for ORT assessment. Moreover, RS and RCI exhibit a high degree of spatial consistency, which explains the similar extent of performance degradation observed when either modality is missing.

2) Results for Africa Dataset: Table V presents a quantitative comparison of the CRFNet model with six other methods on the complete Africa Dataset regarding OA and kappa coefficients, with the optimal results highlighted in bold. Clearly, the results from the Africa Dataset are significantly better than those from the Asia Dataset. A plausible explanation is that the Asia Dataset mainly consists of mountainous and hilly regions with trees as the primary LC, whereas the Africa Dataset mainly consists of plains and plateaus with bare ground as the primary LC. Extensive tree cover can significantly affect the accuracy of ORT assessment, a conclusion that aligns with common understanding. The CRFNet model achieved optimal assessment results on the Africa Dataset, with an average OA of 92.54% and kappa of 86.46%, and standard deviations of 0.09% and 0.15%, respectively. Comparing Tables III and V reveals that in the Asia Dataset, the CRFNet model improved the OA and kappa by 6.66% and 10.74%, respectively, compared to the second-optimal model, FusionMixer, whereas in the Africa Dataset, the OA and kappa improved by 1.2% and 2.06%, respectively, compared to the second-most-optimal model, FusionMixer. This indicates that the proposed CRFNet model has a significant advantage in complex geological environments. In addition, DL methods still possess a substantial accuracy advantage over rule-based methods.

Table VI presents a quantitative comparison of the CRFNet model and other methods on the incomplete Africa Dataset in terms of OA and kappa, with the optimal results highlighted in bold. The experimental strategy was the same as in Table IV, with the test data masked for RS, GHs, and RCI, and the

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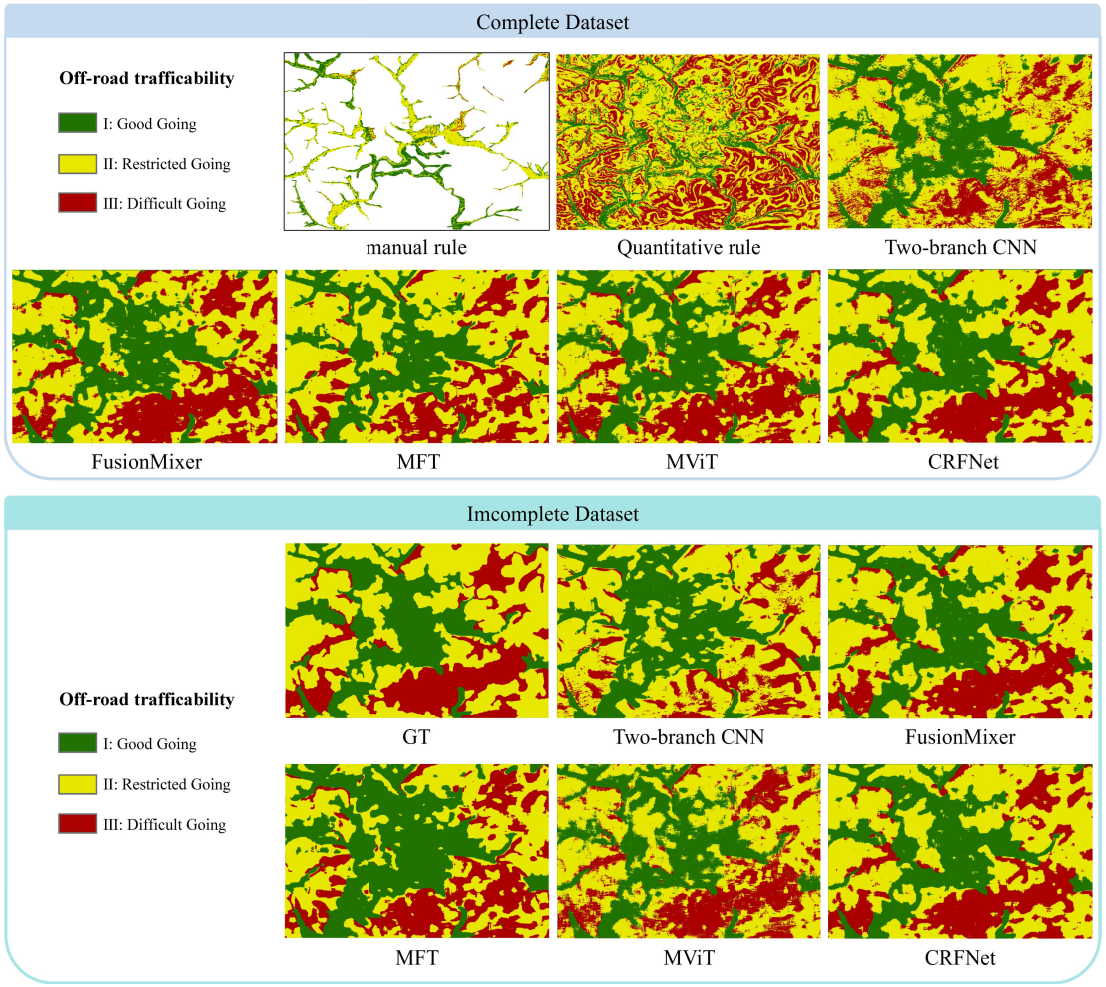


Fig. 5. ORT assessment results of different comparative methods on the Asia Dataset.

models retrained. Comparing Tables III and VI reveals the following: 1) the accuracy of DL methods on the incomplete Africa Dataset is higher than on the complete Asia Dataset, and 2) when multimodal data are missing, the OA and kappa coefficients of all methods significantly decrease, although the accuracy decline is less severe in the incomplete Africa Dataset than in the complete Asia Dataset. This clearly demonstrates the significant impact of geological and geographical environments on ORT assessment. The CRFNet model demonstrated optimal performance on the incomplete Africa Dataset, with average OA and kappa coefficients of 91.95% and 85.34%, respectively, representing decreases of 0.59% and 1.12%. The ExViT model experienced the most severe decline, with OA and kappa coefficients decreasing by 8.26% and 13.1%, respectively. Among all methods, the accuracy for Label III significantly declined, indicating that in the Africa Dataset, the RS, GHs, and RCI factors greatly influenced Label III. In the analysis of the impact of single-modality absence on model performance, the experimental results for the Africa and Asia Datasets are quite similar. In both cases, the absence of the GHs modality causes the least degradation in performance, while the missing RS and RCI modalities lead to more significant declines. Overall, CRFNet consistently outperforms other compared methods under all single-modality missing scenarios, demonstrating superior generalization capability and robustness across different geological environments.

3) Visual Comparison: Figs. 5 and 6 illustrate the visual results of ORT assessments using different comparison methods on the Asia Dataset and the Africa Dataset. Comparing the two ground truth (GT) datasets reveals that the Africa Dataset is predominantly classified as Label 2 (“II”), with an ORT corridor running from the northwest to the southeast within the study area. In contrast, the Asia Dataset exhibits a complex intermingling of the Label 3 (“III”) attribute types, lacking any continuous corridor. Overall, the natural geological and topographical conditions of the Africa Dataset confer a distinct advantage in terms of trafficability.

On the complete dataset, manual rule methods are limited by the assessment factors, making it difficult to directly assess most areas. QRM are significantly superior to manual rule methods; however, the assessment results exhibit many visual misclassifications and a significant amount of salt-and-pepper noise. Adding supplementary information from RSI can effectively reduce the deficiencies in ORT assessment maps. Moreover, due to its superior feature extraction capability, DL effectively learns the complex interactions among input and output feature maps. Consequently, the two-branch CNN generates relatively accurate ORT assessment maps, achieving

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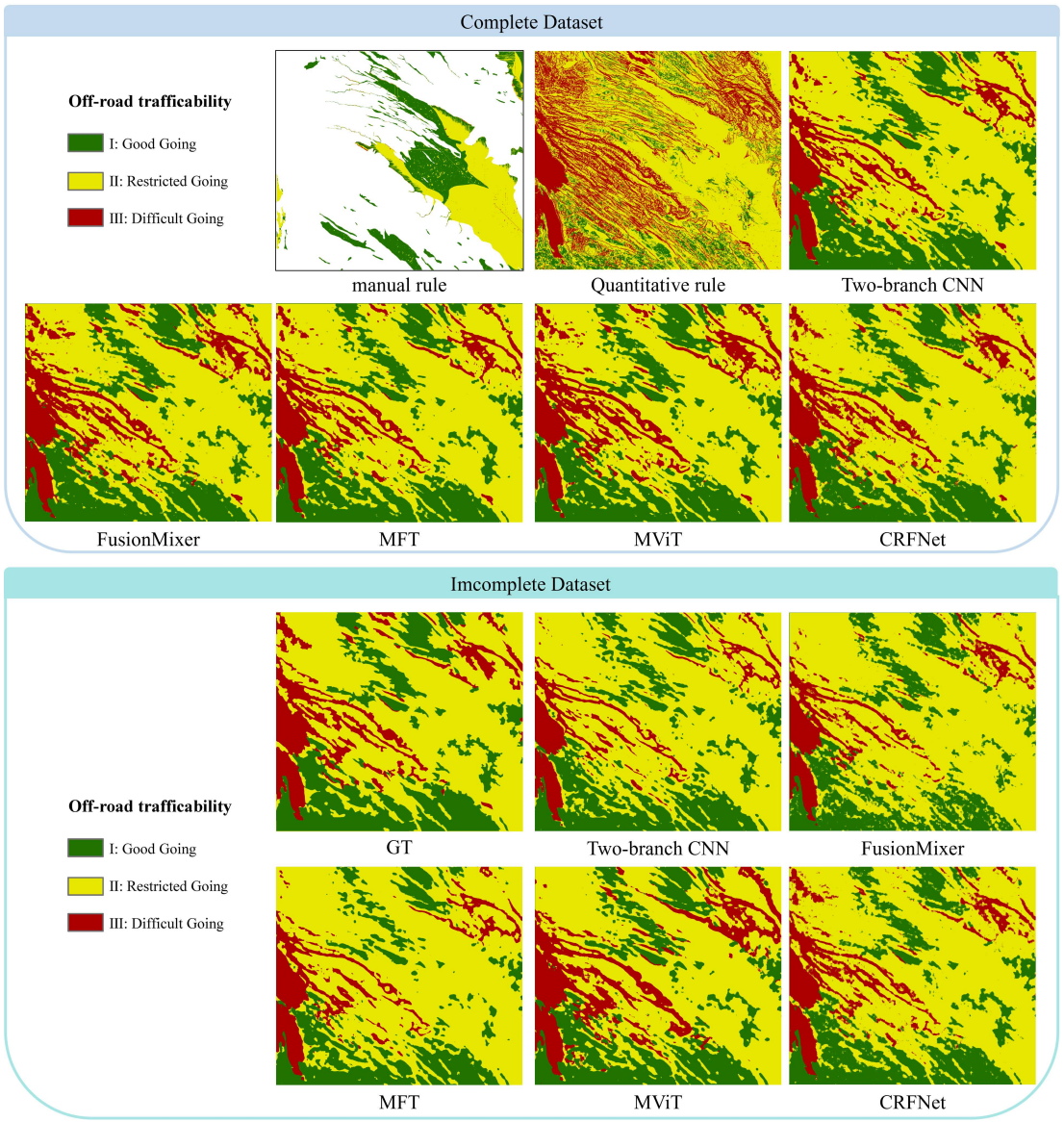


Fig. 6. ORT assessment results of different comparative methods on the Africa Dataset.

good separation for the three types of results. However, compared to other DL methods, the two-branch CNN exhibits more visual misclassifications. The deep feature fusion model enhances the learning ability of the model for nonlinear sequential relationships in the data through targeted feature extraction and efficient feature fusion, resulting in distinct class boundaries and significantly improved visual quality. Due to the multimodal feature correction and better information exchange of RSI and factors, the CRFNet model results show significant improvements. Specifically, the results of the ORT assessment are more realistic and finely detailed.

On the incomplete Asia Dataset, due to the missing of RS, GHs, and RCI data, the FusionMixer, MFT, ExViT, and MAPs models show a large number of significant misclassifications for Label 1 ("I") in the central region. On the incomplete Africa Dataset, a large number of misclassifications for Label 3 ("III") appear in the left region. The MtPLS enriches the multimodal dataset for ORT assessment, and the complementary learning of complete and incomplete datasets enhances its generalization ability and prediction accuracy across more scenarios.

D. Ablation Study

1) Sensitivity Analysis of Model Modules: Ablation experiments were used to validate the effectiveness of the FRM, FFMM, and MtPLS modules in the CRFNet model. Ablation studies were performed using the incomplete dataset, with findings presented in Table VII. The baseline model, the ResNet50 shallow backbone network, extracts RSI and Factor image features using two branches. After simple fusion, the OA and kappa coefficients for the Asia Dataset are 64.24% and 40.50%, respectively, while for the Africa Dataset, these values are 80.85% and 65.00%. When adding a single module, either the FRM or FFMM, both OA and kappa coefficients show significant improvement. It is noteworthy that the inclusion of the FFMM module results in more pronounced improvements in OA and kappa coefficients. When both FRM and FFMM

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TABLE VII
SENSITIVITY ANALYSIS OF CRFNET MODEL MODULES

Method	Dataset	ResNet50 Backbone	FRM	FMM	MtPLS	1	2	3	OA	Kappa
CRFNet	Asia	√				45.95	91.62	33.34	64.24 ± 1.34	40.50 ± 2.76
		√	√			52.09	84.78	81.26	74.10 ± 0.69	55.83 ± 1.26
		√		√		47.78	**92.90**	81.08	76.09 ± 1.40	59.30 ± 2.98
		√	√	√		76.01	92.66	83.79	86.01 ± 0.06	77.66 ± 0.04
		√	√	√	√	**80.91**	89.55	**87.47**	**87.16 ± 0.15**	**79.45 ± 0.27**
	Africa	√				80.77	79.00	86.18	80.85 ± 0.93	65.00 ± 1.86
		√	√			88.07	83.77	73.34	82.24 ± 0.14	71.76 ± 6.34
		√		√		86.43	82.56	**86.76**	85.22 ± 0.60	73.44 ± 0.92
		√	√	√		90.00	94.22	75.60	90.28 ± 0.93	82.16 ± 1.86
		√	√	√	√	**91.40**	**95.17**	79.04	**91.95 ± 0.10**	**85.34 ± 0.21**

TABLE VIII
SENSITIVITY ANALYSIS OF FRM, FMM, AND MTPLS VARIANTS IN CRFNET

Feature Rectification	Feature Fusion	Modality Missing	Asia		Africa	
			OA	Kappa	OA	Kappa
channel-wise *only*	FMM	MtPLS	86.39	78.46	90.65	84.19
spatial-wise *only*	FMM	MtPLS	86.30	78.42	90.59	84.07
FRM	token-mixing-MLP *only*	MtPLS	86.91	79.21	91.69	84.92
FRM	channel-mixing-MLP *only*	MtPLS	86.89	78.91	91.27	84.79
FRM	FMM	incomplete data *only*	86.81	78.88	91.37	84.84
FRM	FMM	prompt vectors *only*	86.93	79.34	91.74	85.13
FRM	FMM	MtPLS	**87.16**	**79.45**	**91.95**	**85.34**

modules are added, the OA and kappa coefficients are 86.01% and 77.66% for the Asia Dataset, and 90.28% and 82.16% for the Africa Dataset, respectively. After adding the MtPLS module, the OA and kappa coefficients increased by 1.15% and 1.79% for the Asia Dataset, and by 1.67% and 3.18% for the Africa Dataset, respectively. Among the three ORT classes, III in the Asia Dataset shows the poorest performance under the baseline model, with an accuracy of only 33.34%. This suggests that models lacking deep cross-modal modeling capabilities struggle to accurately identify regions classified as "Difficult Going." As the FRM, FFMM, and MtPLS modules are progressively introduced, the classification accuracy for all three classes steadily improves, with "III" showing the most significant gains. These results further demonstrate that the FRM module effectively rectifies modality discrepancies between RSI and Factors data, the FFMM module efficiently fuses the rectified features, and the MtPLS module enriches the representation of incomplete modalities, thereby enhancing the generalization ability of the CRFNet model.

2) Sensitivity Analysis of Module Variants: To comprehensively analyze the individual contributions of each component within CRFNet, we designed several model variants to evaluate the impact of the FRM, FFMM, and MtPLS modules on overall performance. These variants include: 1) using only channel-wise rectification in the FRM module; 2) using only spatial-wise rectification in the FRM module; 3) using only the token-mixing MLP in the FFMM module; 4) using only the channel-mixing MLP in the FFMM module; 5) using only incomplete data as input in the MtPLS module; and 6) using only prompt vectors in the MtPLS module. By comparing the OA and Kappa performance of these variants, we aim to elucidate the specific role each module plays in ORT assessment. The experimental results are summarized in Table VIII. For the FRM module, the variant using only channel-wise rectification slightly outperformed the one using only spatial-wise rectification, indicating that channel-wise information plays a more crucial role in cross-modal feature rectification. In the FFMM module, the token-mixing MLP-only variant performed better than the channel-mixing MLP-only variant, highlighting the importance of token-level feature interaction for effective multimodal fusion, while channel-mixing further enhances the representational capacity of the fused features. For the MtPLS module, the variant using only prompt vectors achieved higher accuracy than the variant using only incomplete data, suggesting that the incorporation of prompt vectors significantly improves the model's generalization ability under modality-missing conditions.

3) Analysis of Image Patch Size: The ORT of a pixel is influenced not only by RSI-factors but also by the attributes of the surrounding pixels. For instance, a pixel rated as "I" should be considered "III" if it is surrounded by multiple "III" pixels. Therefore, analyzing the size of the input image patch is worthwhile when assessing the ORT of a pixel. This work specifically analyzed the correlation among image patch size (r), OA, and runtime across two datasets. Here, an area of $r \times r$ encapsulates the spatial characteristics of the central pixel. As shown in Fig. 7, the Asia Dataset and Africa Dataset achieve optimal ORT evaluation performance at $r = 17$ and $r = 19$, respectively. This phenomenon can be explained by the differences in spatial structures between the two datasets. The

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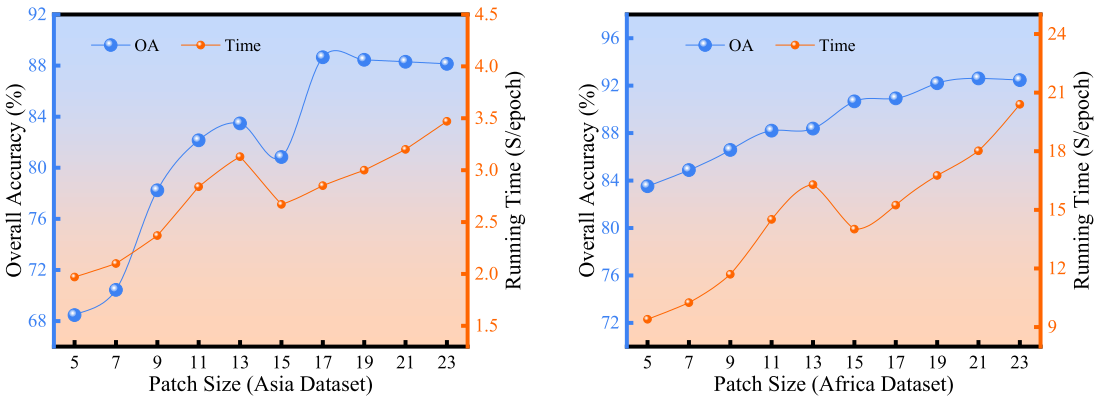


Fig. 7. OA and running time per epoch on varying sizes of the image patch.

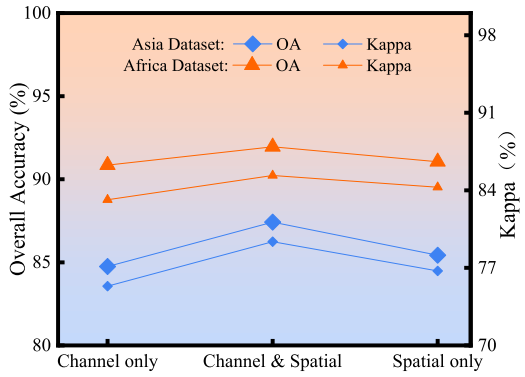


Fig. 8. Ablation experiments on the λ_c and λ_s hyperparameters in FRM.

NE dataset, dominated by vegetation-rich mountains and hills, exhibits higher spatial heterogeneity and stronger regionalized variables, leading to a prominent nugget effect, where abrupt changes dominate at smaller scales. As a result, a smaller patch size suffices to capture key local variations. In contrast, the Djibouti dataset, characterized by exposed plateaus and plains, exhibits weaker spatial variability, reducing the nugget effect and causing the variogram to stabilize over a larger range. Consequently, a larger patch size is required to encompass broader contextual information. This observation aligns with the theory of regionalized variables in geostatistics, which posits that different terrains exhibit varying spatial continuity and variability.

4) Hyperparameter Analysis of the Model: FRM includes channel-wise and spatial-wise feature rectification, where λ_c and λ_s are hyperparameters balancing these two channels. Fig. 8 illustrates the correlation between the λ_c and λ_s hyperparameters and the OA and kappa coefficients. "channel only" refers to using only channel-wise feature rectification (i.e., $\lambda_c = 1$ and $\lambda_s = 0$). The OA and kappa coefficients are 84.75% and 75.35% for the Asia Dataset, and 90.85% and 83.15% for the Africa Dataset, respectively. "spatial only" indicates using only spatial-wise feature rectification (i.e., $\lambda_c = 0$ and $\lambda_s = 1$). The OA and kappa coefficients for the Asia Dataset are 85.43% and 76.72%, respectively, while for the Africa Dataset, they are 91.5% and 84.98%, respectively. "Channel and spatial" refers to both channel-wise and spatial-wise feature rectification. The OA and kappa coefficients are 87.16% and 79.45% for the Asia Dataset, and 91.95% and 85.34% for the Africa Dataset, respectively. As shown in the figure, it is evident that relying on either channel-wise or spatial-wise rectification alone results in suboptimal performance on both datasets. Channel-wise rectification captures interchannel dependencies and adjusts the importance of each channel, enabling the model to focus on the most relevant features for the ORT assessment task. Spatial-wise rectification optimizes local spatial information by considering the spatial distribution of features, which is particularly effective in regions with high spatial heterogeneity (e.g., mountainous terrains). Combining these two mechanisms produces a synergistic effect, enhancing the overall robustness and accuracy of the CRFNet model. When λ_c and λ_s are both set to their default values of 0.5, the CRFNet model achieves optimal ORT assessment performance on both datasets.

In multitask learning, λ_{com} and λ_{mis} are hyperparameters that balance the loss between the complete and incomplete datasets. Fig. 9(a) illustrates the relationship between different λ values and OA and kappa coefficients. When $\lambda_{\text{com}} = 1$ and $\lambda_{\text{mis}} = 0$, the CRFNet model is trained using the complete dataset. The OA and kappa coefficients for the Asia Dataset are 80.35% and 67.69%, respectively, while for the Africa Dataset, they are 84.94% and 72.16%, respectively. When $\lambda_{\text{com}} = 0$ and $\lambda_{\text{mis}} = 1$, the CRFNet model is trained using the incomplete dataset. The OA and kappa coefficients for the Asia Dataset are 84.36% and 74.83%, respectively, while for the Africa Dataset, they are 88.46% and 78.49%, respectively. When $\lambda_{\text{com}} > 0$ and $\lambda_{\text{mis}} > 0$, the CRFNet model uses both the complete and incomplete datasets for multitask learning. When both hyperparameters are set to their default values of 0.5, the OA and kappa coefficients for the Asia Dataset are 87.16% and 79.45%, respectively, while for the Africa Dataset, they are 91.95% and 85.34%, respectively, achieving the optimal ORT assessment. This demonstrates that single-task learning results in suboptimal CRFNet model performance. Missing modalities reduce the model's ability to leverage complementary information from multiple data sources (e.g., RS, GHs, and RCI), thereby limiting its understanding of terrain features. In addition, the patterns of missing data are often irregular and unpredictable, leading to higher dataset variability. By simultaneously utilizing both the complete and incomplete datasets, the model ensures it can capture detailed

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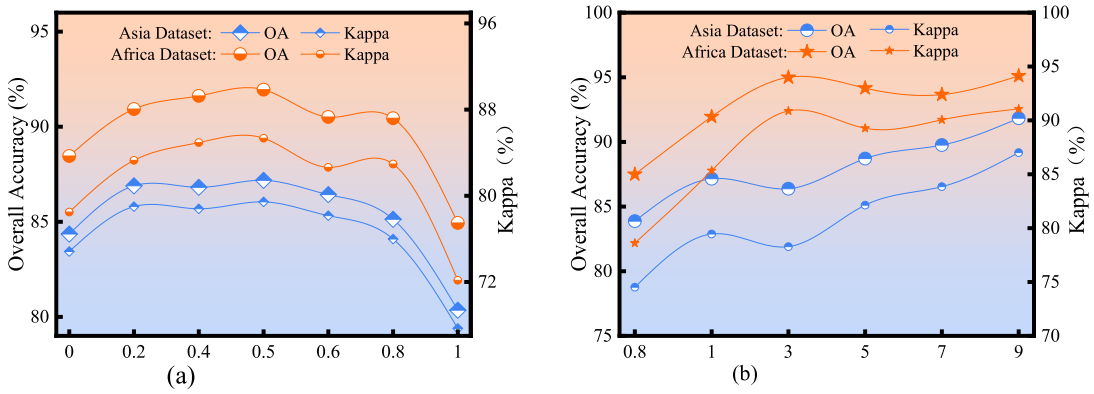


Fig. 9. Ablation experiments of the hyperparameters of the CRFNet model. (a) λ_{com} or $(1-\lambda_{mis})$. (b) Training size (%).

TABLE IX
QUANTITATIVE COMPARISON OF MODEL COMPLEXITY IN TERMS OF PARAMETERS AND FLOPS

Method	CNN	Two-branch CNN	FusionMixer	MFT	ExViT	CRFNet
PARAMETERS (M)	0.36	1.09	7.99	0.22	0.23	3.73
FLOPs (G)	0.004	0.0059	0.16	0.0028	3.51	0.0031

terrain information (through the complete dataset) while also adapting to real-world incomplete data scenarios.

5) Performance Over Varying Training Ratio: Evaluation of the model's performance relative to the number of training samples is conducted to check whether the CRFNet model experiences underfitting or overfitting when training samples are insufficient or unreliable. Consequently, 0.8%, 1%, 3%, 5%, 7%, and 9% of the training samples were arbitrarily chosen for model training, with other parameters remaining unchanged. Fig. 9(b) illustrates the correlation between different training sample sizes and the OA and kappa coefficients. In both datasets, the OA and kappa coefficients exhibit a similar pattern, initially rising, then falling, then rising again. An extremum for the Asia Dataset occurs at 1% of the training samples, while for the Africa Dataset, the first extremum appears at 3% of the training samples. ORT assessment samples undergo noise filtering, boundary smoothing, and manual calibration, making the sample labeling process time-consuming and prone to errors. Therefore, a balance between sample size and accuracy is needed. In this work, we chose 1% of the labels as the training samples for both datasets.

6) Complexity Analysis of the CRFNet Model: In practical applications, besides OA and kappa coefficients, the efficiency of DL models is also a crucial consideration, typically including costs in terms of time and space. Common quantitative comparison metrics encompass the number of model parameters and floating-point operations (FLOPs). Table IX quantitatively compares the complexity of the CNN, two-branch CNN, FusionMixer, MFT, ExViT, and CRFNet using parameters and FLOPs as metrics. As shown in the table, the FusionMixer model based on the ResNet network has the most parameters due to its deeper architecture and hierarchical feature interaction, inevitably resulting in more parameters and computations. The MFT model based on the Transformer network has the lowest complexity in terms of parameters and FLOPs, thanks to shared weights and the hierarchical attention mechanism. In contrast, the ExViT model, also based on the Transformer network, reduces the number of parameters through depthwise separable convolutions and subspace projections; however, its repeated sublayer structure and extensive use of multihead attention result in relatively high FLOPs. Although ExViT excels in learning multimodal features, it requires significant computational resources. Note that the CRFNet demonstrates a moderate level of parameter and FLOPs complexity among the six models, achieving an optimal tradeoff between efficiency and performance.

V. CONCLUSION

In this work, we propose a CRFNet model that integrates RSI-factors data to assess ORT. Accurate and intelligent assessment of ORT is essential for planning operations and route navigation in scenarios such as military actions, emergency rescue, and disaster response. The CRFNet model effectively leverages the complementary information provided by RSI-factors data. It incorporates a ResNet50 shallow backbone, FRM, FFMM, and MtPLS modules, enabling accurate and intelligent ORT assessment. Experiments conducted in two study regions validate the model's superiority and practicality. The CRFNet model addresses these challenges by introducing the FRM, which utilizes spatial, channel, and interaction correlations to reduce uncertainty and noise in RSI-factors data. The FFMM enables adaptive cross-fusion of spectral images with geographical and geological environmental factors, fully leveraging the cross-modal information of RSI-factors. To address potential missing data in factors within complex geological environments, the MtPLS model generates modal incomplete data to enrich RSI-factors data types, adding input-level prompts for specific missing data types, thereby improving the performance of the CRFNet model.

Despite the effectiveness of CRFNet in addressing key challenges associated with ORT assessment, certain limitations remain. Although the model demonstrates robustness to incomplete auxiliary data, its performance may degrade when auxiliary inputs are extensively missing or of low quality—particularly in cases where geological information is unavailable or geographic features are provided at coarse spatial resolutions. As such, the model's effectiveness remains dependent on the reliability and resolution of auxiliary modalities. In addition, the current architecture lacks support for dynamic modality adaptation and hierarchical multiscale feature integration, limiting its scalability to more diverse and

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complex terrain conditions. The reliance on low-resolution data for estimating critical parameters such as surface moisture and RCI also introduces inherent uncertainty. To improve robustness and generalization, future work will explore the integration of Transformer-based modules to better capture long-range dependencies in complex terrain environments. We also plan to investigate transfer learning strategies, including pretraining in data-rich regions and fine-tuning with limited or unlabeled samples in the target domain, to enhance cross-domain adaptability. Furthermore, our current annotation efforts primarily focus on hilly and plain areas, which represent common off-road scenarios. However, the lack of labeled datasets for more challenging environments—such as deserts—remains a critical limitation. To address this, we will conduct more extensive field surveys to improve the accuracy, diversity, and representativeness of ORT ground-truth datasets.

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